

EXPERIMENT-3

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1.COMMAND LINE INTERFACE:

PYTHON CODE:

```
import os

import sys

def rename_file(old_name, new_name):

    try:

        os.rename(old_name, new_name)

        print(f"File renamed from {old_name} to {new_name}")

    except FileNotFoundError:

        print(f"Error: {old_name} not found.")

    except Exception as e:

        print(f"An error occurred: {e}")

if __name__ == "__main__":

    if len(sys.argv) != 3:

        print("Usage: python rename_file_cli.py <old_filename> <new_filename>")

    else:

        rename_file(sys.argv[1], sys.argv[2])
```

OUTPUT:

```
PS C:\Users\Adhithya\Documents\UID> python file_rename.py old.txt new.txt
File renamed from old.txt to new.txt
PS C:\Users\Adhithya\Documents\UID>
```

2.GRAPHICAL USER INTERFACE:

PYTHON CODE:

```
import tkinter as tk

from tkinter import messagebox

import os


def rename_file():

    old_name = old_filename_entry.get().strip()

    new_name = new_filename_entry.get().strip()


    if not old_name or not new_name:

        messagebox.showwarning("Warning", "Please enter both filenames!")

        return


    if not os.path.exists(old_name):

        messagebox.showerror("Error", f"File '{old_name}' not found.")

        return


    if os.path.exists(new_name):

        overwrite = messagebox.askyesno("Warning", f"'{new_name}' already exists. Overwrite?")

        if not overwrite:

            return


    try:

        os.rename(old_name, new_name)

        messagebox.showinfo("Success", f"File renamed from '{old_name}' to '{new_name}'")

    except Exception as e:

        messagebox.showerror("Error", f"An error occurred: {e}")
```

```

root = tk.Tk()

root.title("File Renamer")

root.geometry("400x150")

root.resizable(False, False)

tk.Label(root, text="Old Filename:", font=("Arial", 10)).grid(row=0, column=0, padx=10, pady=5,
sticky="w")

tk.Label(root, text="New Filename:", font=("Arial", 10)).grid(row=1, column=0, padx=10, pady=5,
sticky="w")

old_filename_entry = tk.Entry(root, width=40)

old_filename_entry.grid(row=0, column=1, padx=10, pady=5)

new_filename_entry = tk.Entry(root, width=40)

new_filename_entry.grid(row=1, column=1, padx=10, pady=5)

rename_button = tk.Button(root, text="Rename File", bg="green", fg="white", font=("Arial", 10, "bold"),
command=rename_file)

rename_button.grid(row=2, column=0, columnspan=2, pady=10)

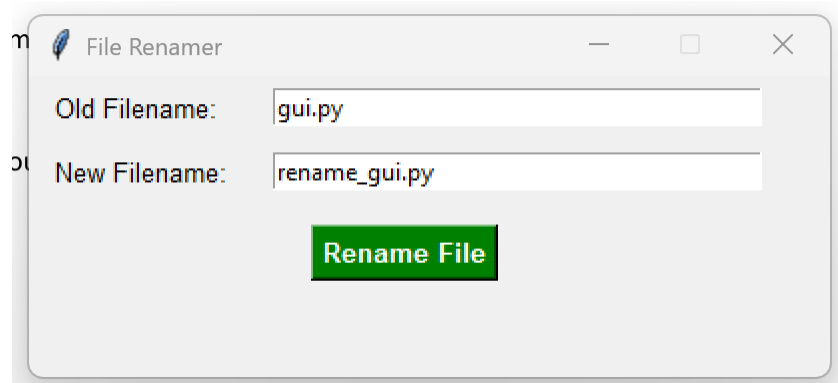
root.mainloop()

```

```

PS C:\Users\Adhithya\Documents\UID> python rename_gui.py

```



3.VOICE USER INTERFACE:

PYTHON CODE:

```
import speech_recognition as sr
```

```
import os
```

```
def rename_file_from_voice_command(command):
```

```
    try:
```

```
        words = command.lower().split(" ")
```

```
        if "rename" in words and "to" in words:
```

```
            rename_index = words.index("rename")
```

```
            to_index = words.index("to")
```

```
            old_name = words[rename_index + 1]
```

```
            new_name = words[to_index + 1]
```

```
            if not os.path.exists(old_name):
```

```
                print(f"Error: File '{old_name}' not found.")
```

```
                return
```

```
            os.rename(old_name, new_name)
```

```
            print(f" File renamed from '{old_name}' to '{new_name}'")
```

```
        else:
```

```
            print("Invalid command format. Say: 'Rename oldfile.txt to newfile.txt'")
```

```
    except Exception as e:
```

```
        print(f"Error: {e}")
```

```
def listen_for_command():
```

```
    recognizer = sr.Recognizer()
```

```
    mic = sr.Microphone()
```

```
    print(" Listening for command to rename a file...")
```

```

with mic as source:

    recognizer.adjust_for_ambient_noise(source)

    audio = recognizer.listen(source)

    try:

        command = recognizer.recognize_google(audio)

        print(f" Command received: {command}")

        rename_file_from_voice_command(command)

    except sr.UnknownValueError:

        print(" X Sorry, I couldn't understand the command.")

    except sr.RequestError as e:

        print(f"A Could not request results from Google Speech Recognition service; {e}")

if __name__ == "__main__":

    listen_for_command()

```

```

PS C:\Users\Adhithya\Documents\UID> python vui.py
Listening for command to rename a file...
Command received: rename
Invalid command format. Say: 'Rename oldfile.txt to newfile.txt
PS C:\Users\Adhithya\Documents\UID> python vui.py
Listening for command to rename a file...
X Sorry, I couldn't understand the command.
PS C:\Users\Adhithya\Documents\UID> |

```

```

PS C:\Users\Adhithya\Documents\UID> python vui.py
Listening for command to rename a file...
Command received: rename a.txt to new.txt
File renamed from 'a.txt' to 'new.txt'

```

4.USER SATISFACTION COMPARISON:

PYTHON CODE:

```
def survey():

    print("Rate your satisfaction with the following interfaces (1-5):")

    try:

        cli_satisfaction = int(input("CLI (1-5): "))

        gui_satisfaction = int(input("GUI (1-5): "))

        vui_satisfaction = int(input("VUI (1-5): "))

        if not (1 <= cli_satisfaction <= 5 and 1 <= gui_satisfaction <= 5 and 1 <= vui_satisfaction <= 5):

            print("Please enter ratings between 1 and 5 only.")

            return

        print("\nYour satisfaction ratings:")

        print(f"CLI: {cli_satisfaction}")

        print(f"GUI: {gui_satisfaction}")

        print(f"VUI: {vui_satisfaction}")

        avg_satisfaction = (cli_satisfaction + gui_satisfaction + vui_satisfaction) / 3

        print(f"\nAverage Satisfaction Score: {avg_satisfaction:.2f}")

    except ValueError:

        print("Invalid input! Please enter numbers between 1 and 5.")

if __name__ == "__main__":

    survey()
```

```
PS C:\Users\Adhithya\Documents\UID> python comparison.py
Rate your satisfaction with the following interfaces (1-5):
CLI (1-5): 4
GUI (1-5): 5
VUI (1-5): 2

Your satisfaction ratings:
CLI: 4
GUI: 5
VUI: 2

Average Satisfaction Score: 3.67
PS C:\Users\Adhithya\Documents\UID> |
```